

13 — Final Architecture Review

Purpose

The purpose of this document is to perform a final evaluation of the complete Mosaica Architecture.

This module reviews the overall system design and confirms that all architectural components work together as a coherent, maintainable and development-ready system.

The objective is to verify that Mosaica has:

- consistent architecture,
 - complete design foundations,
 - clear system boundaries,
 - readiness for implementation.
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Review Scope

This final review evaluates the complete Mosaica Architecture, including:

- Product Foundation,
 - Business Rules,
 - Domain Model,
 - Database Specification,
 - API Specification,
 - Application Architecture,
 - Security Architecture,
 - Infrastructure Architecture,
 - Development Architecture,
 - User Experience Architecture,
 - Future Capability Architecture,
 - Product Evolution Architecture.
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Final Review Principles

Principle 1 — Architectural Consistency

All system modules must work together without conflicting decisions.

Principle 2 — Completeness Before Development

Major architectural decisions should be defined before implementation begins.

Principle 3 — Preserve Simplicity

The architecture should remain as simple as possible while satisfying product requirements.

Principle 4 — Protect Core Vision

Technical and product decisions must continue supporting Mosaica's original purpose.

Principle 5 — Controlled Transition to Development

Architecture completion should provide a clear foundation for implementation.

Review Structure

13 — Final Architecture Review

13.01 Architecture Consistency Review

13.02 Completeness Review

13.03 Development Readiness Review

13.04 Final Architecture Approval

13.01 — Architecture Consistency Review

Purpose

The purpose of this section is to validate the consistency of the complete Mosaica Architecture.

This review evaluates whether all architectural modules work together as a unified system.

The objective is to confirm that architectural decisions:

- do not conflict with each other,
 - preserve system boundaries,
 - support the original product vision.
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Review Scope

The following architecture modules are reviewed:

- 01 Product Foundation
 - 02 Business Rules
 - 03 Domain Model
 - 04 Database Specification
 - 05 API Specification
 - 06 Application Architecture
 - 07 Security Architecture
 - 08 Infrastructure Architecture
 - 09 Development Architecture
 - 10 User Experience Architecture
 - 11 Future AI Capability Architecture
 - 12 Product Evolution Architecture
-

Consistency Review Areas

1. Product Vision Alignment

Status: Approved

The architecture remains aligned with the original Mosaica vision.

Confirmed:

Mosaica continues to focus on:

- discovering cultural content,
- building a personal archive,

- recording personal experiences,
 - creating long-term cultural memory.
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2. Domain and Business Alignment

Status: Approved

The Domain Model correctly represents the business rules.

Confirmed:

Business decisions remain inside the Domain Layer.

The system avoids placing business logic inside:

- UI,
 - API,
 - Database,
 - external services.
-

3. Database Alignment

Status: Approved

The Database Specification supports the Domain Model.

Confirmed:

Database structures represent system concepts without replacing business meaning.

Relationship:

Domain Concepts

↓

Data Representation

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Database Structure

4. API Alignment

Status: Approved

The API architecture correctly exposes system capabilities.

Confirmed:

API responsibilities remain:

- communication,
- request processing,
- response delivery.

The API does not contain core business rules.

5. Application Layer Alignment

Status: Approved

The Application Layer correctly coordinates system operations.

Confirmed:

The Application Layer connects:

User Request

↓

Application Workflow

↓

Domain Operation

6. Security Alignment

Status: Approved

Security decisions are integrated across the architecture.

Confirmed:

Security is considered in:

- data handling,

- authentication,
 - authorization,
 - infrastructure,
 - development practices.
-

7. Infrastructure Alignment

Status: Approved

Infrastructure decisions support Mosaica's current development approach.

Confirmed:

The system does not require:

- expensive infrastructure,
 - mandatory external services,
 - unnecessary operational complexity.
-

8. Development Alignment

Status: Approved

Development practices correctly support architectural decisions.

Confirmed:

The following remain aligned:

- repository structure,
 - coding standards,
 - testing approach,
 - controlled deployment.
-

9. UX Alignment

Status: Approved

The User Experience Architecture correctly represents Mosaica's purpose.

Confirmed:

The experience follows:

Discover

↓

Collect

↓

Experience

↓

Record

↓

Reflect

10. Future Capability Alignment

Status: Approved

Future capabilities remain separated from the core system.

Confirmed:

AI and advanced features are optional.

The initial system remains functional without:

- AI,
 - paid services,
 - external dependencies.
-

Architecture Boundary Validation

Confirmed system boundary:

User Interface

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Application Layer

↓
Domain Layer

↓
Infrastructure Layer

↓
External Systems

No layer bypasses another layer's responsibility.

Complexity Review

Status: Approved

The architecture avoids unnecessary complexity.

Confirmed:

Mosaica does not require:

- enterprise-level infrastructure,
 - expensive tools,
 - premature automation.
-

Consistency Review Conclusion

The Mosaica Architecture is internally consistent.

No major conflicts have been identified between architectural modules.

The current architecture provides a coherent foundation for future development.

13.02 — Completeness Review

Purpose

The purpose of this section is to validate whether the Mosaica Architecture contains all essential foundations required for implementation.

This review evaluates whether critical architectural areas have been defined sufficiently before development begins.

The objective is to confirm that no major architectural responsibility has been overlooked.

Completeness Review Scope

The review evaluates:

Product Definition

Business Logic

Domain Structure

Data Management

System Communication

Application Behavior

Security

Infrastructure

Development Process

User Experience

Future Evolution

Completeness Evaluation

1. Product Foundation

Status: Complete

The product purpose and scope have been defined.

Confirmed:

- product identity,
- target experience,
- core value,
- initial boundaries.

2. Business Rules

Status: Complete

The fundamental business logic has been documented.

Confirmed:

- important rules,
- system behavior,
- domain expectations.

3. Domain Model

Status: Complete

The core business concepts have been defined.

Confirmed:

- entities,
- relationships,
- domain responsibilities.

4. Database Architecture

Status: Complete

The data foundation has been specified.

Confirmed:

- data organization,
 - persistence approach,
 - relationship principles.
-

5. API Architecture

Status: Complete

System communication boundaries have been defined.

Confirmed:

- external communication principles,
 - responsibility separation,
 - integration approach.
-

6. Application Architecture

Status: Complete

Application workflows and internal coordination have been defined.

Confirmed:

- layer responsibilities,
 - use-case approach,
 - workflow organization.
-

7. Security Architecture

Status: Complete

Security foundations have been established.

Confirmed:

- data protection principles,
 - access considerations,
 - security responsibilities.
-

8. Infrastructure Architecture

Status: Complete

Infrastructure requirements have been considered.

Confirmed:

- environment approach,
 - deployment philosophy,
 - cost awareness.
-

9. Development Architecture

Status: Complete

Development practices have been defined.

Confirmed:

- repository structure,
 - coding standards,
 - testing approach,
 - CI/CD philosophy.
-

10. User Experience Architecture

Status: Complete

User interaction foundations have been established.

Confirmed:

- navigation,
 - user flows,
 - interface principles.
-

11. Future Capability Architecture

Status: Complete

Future AI capabilities have been separated from the core product.

Confirmed:

- AI is optional,
 - core system does not depend on AI,
 - future expansion requires evaluation.
-

12. Product Evolution Architecture

Status: Complete

Future growth principles have been established.

Confirmed:

- feature evaluation,
 - version strategy,
 - change management.
-

Missing Architecture Areas Evaluation

The following areas were evaluated:

Payment Architecture

Status: Not Required

Reason:

The initial Mosaica scope does not require commercial monetization.

Future business models require separate evaluation.

Multi-User Collaboration Architecture

Status: Not Required

Reason:

Initial product focus is personal archive usage.

Advanced Analytics Architecture

Status: Future Capability

Reason:

Useful potential exists, but it is not required for initial implementation.

AI Implementation Architecture

Status: Future Capability

Reason:

AI is intentionally excluded from the initial product foundation.

Enterprise Operations Architecture

Status: Not Required

Reason:

Mosaica is not designed as an enterprise system at this stage.

Completeness Conclusion

The Mosaica Architecture contains all critical foundations required to begin implementation.

Remaining undefined areas are either:

- future capabilities,
- optional expansions,
- or outside the current product scope.

No blocking architectural gaps have been identified.

Architectural Decisions Confirmed

Decision 01

The architecture is sufficiently complete for initial development.

Decision 02

Not every possible future capability requires present-day architecture.

Decision 03

Optional systems remain separate from the core product.

Decision 04

Future expansion areas can be addressed through later decisions.

13.03 — Development Readiness Review

Purpose

The purpose of this section is to evaluate whether the Mosaica Architecture provides a sufficient foundation for beginning software development.

This review confirms that the system design, development approach and product boundaries are clear enough to transition from planning into implementation.

The objective is to ensure that development begins with:

- clear direction,
- controlled scope,
- appropriate priorities,
- sustainable foundations.

Development Readiness Scope

The following areas are evaluated:

Product Clarity

Architecture Stability

Development Foundation

Technical Preparation

Scope Control

Implementation Planning

Readiness Evaluation

1. Product Clarity

Status: Approved

Mosaica's purpose and initial scope are clearly defined.

Confirmed:

The initial product focuses on:

- personal cultural archive,
 - content organization,
 - personal experiences,
 - long-term cultural memory.
-

2. Architecture Stability

Status: Approved

The core architecture has been reviewed and stabilized.

Confirmed:

Major system boundaries are defined:

User Interface

↓

Application

↓

Domain

↓

Infrastructure

3. Development Foundation

Status: Approved

Development practices have been established.

Confirmed:

The project has defined:

- repository structure,
 - coding principles,
 - testing approach,
 - version strategy.
-

4. Technical Preparation

Status: Approved

The architecture provides enough technical direction to begin implementation.

Confirmed:

The project does not require before starting:

- paid infrastructure,
 - AI services,
 - external commercial systems.
-

5. Scope Control

Status: Approved

The initial development scope remains controlled.

Excluded from initial implementation:

AI Features

Advanced Analytics

Community Features

Paid Services

Complex Integrations

These remain future possibilities.

6. Development Priority Evaluation

The recommended implementation order is:

Foundation

↓

Core Features

↓

User Experience

↓

Improvements

↓

Optional Expansion

Initial Development Focus

The first implementation phase should prioritize:

Core Data

Creating the fundamental content and archive structures.

Library Experience

Allowing Users to manage personal archive entries.

Personal Organization

Supporting collections and personal structures.

Basic User Experience

Providing a usable interaction flow.

Development Constraints

The initial development should respect:

Cost Constraint

The system should remain possible with free or low-cost tools.

Complexity Constraint

Only required components should be introduced.

Architecture Constraint

New development should respect approved architectural boundaries.

Implementation Readiness Model

Architecture Complete

↓

Development Plan Ready

↓

First Implementation

↓

Validation

↓

Iteration

Development Start Criteria

Mosaica is ready to begin development when:

- ✓ Product goal is clear
 - ✓ Core architecture exists
 - ✓ Initial scope is defined
 - ✓ Development approach is known
 - ✓ Future complexity is controlled
-

Readiness Conclusion

The Mosaica Architecture provides sufficient foundation for beginning implementation.

The project can transition from architecture planning into development.

Future details can be refined during implementation without requiring redesign of the entire system.

Architectural Decisions Confirmed

Decision 01

Development may begin with the current architecture.

Decision 02

Initial implementation should focus on core value.

Decision 03

Future capabilities remain outside the first development scope.

Decision 04

Implementation decisions should respect existing architecture.

13.04 — Final Architecture Approval

Purpose

The purpose of this section is to provide the final approval of the complete Mosaica Architecture.

This approval confirms that the architecture provides a complete, consistent and sustainable foundation for implementation.

The objective is to formally transition Mosaica from architecture design into development.

Final Architecture Review Summary

The following architecture areas have been completed and reviewed:

01 Product Foundation

02 Business Rules

03 Domain Model

04 Database Specification

05 API Specification

06 Application Architecture

07 Security Architecture

08 Infrastructure Architecture

09 Development Architecture

10 User Experience Architecture

11 Future AI Capability Architecture

12 Product Evolution Architecture

Final Validation Results

Product Foundation

Status: Approved

Mosaica's purpose, scope and identity are clearly defined.

The product remains focused on:

- personal cultural archive,
 - meaningful organization,
 - long-term cultural memory.
-

System Architecture

Status: Approved

The system architecture provides clear separation between responsibilities.

Confirmed:

Interface

↓

Application

↓

Domain

↓

Infrastructure

Data and Business Alignment

Status: Approved

The data structure and business concepts remain aligned.

The system represents meaningful concepts rather than only technical storage.

Security Foundation

Status: Approved

Security principles are integrated into the architecture.

Confirmed:

- data protection,
 - ownership control,
 - responsible handling.
-

Development Foundation

Status: Approved

The project has sufficient development guidance.

Confirmed:

- repository organization,
 - coding standards,
 - testing principles,
 - version management.
-

User Experience Foundation

Status: Approved

The User experience architecture supports the intended Mosaica journey.

Confirmed:

Discover

↓

Collect

↓

Experience

↓

Record

↓

Reflect

Future Capability Management

Status: Approved

Future capabilities are separated from the initial product.

Confirmed:

AI and advanced systems are:

- optional,
 - independent,
 - evaluated when needed.
-

Cost and Complexity Validation

Status: Approved

The architecture respects the original development constraints.

Confirmed:

Mosaica can begin development without requiring:

- expensive subscriptions,
 - paid infrastructure,
 - AI services,
 - unnecessary external systems.
-

Final Architecture Decision

Decision

The Mosaica Architecture is approved as the foundation for implementation.

Consequence

Development may begin according to the defined architecture.

Future changes shall follow:

- change management principles,
 - version strategy,
 - architectural decision records.
-

Architecture Status

Mosaica Architecture

Version: Initial Architecture

Status: APPROVED

State: DEVELOPMENT READY

Final Approval Record

Project: Mosaica

Architecture Phase: Completed

Approval: Granted

Next Phase: Development

Architectural Completion Statement

The Mosaica Architecture has established:

- what the system is,
- how it works,
- how it should be developed,
- how it should evolve.

The architecture provides a stable foundation while preserving flexibility for future growth.